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Task 1 – Load the Data into ChatGPT & add the main prompt (role, instruction, context, output indicator)

Prompt:

Act as an expert Data Analyst with specialization in Python and generates high-impact data visualization. I have attached a dataset. Your goal is to help me perform specific analysis tasks, which I will outline after you acknowledge this message.

Requirements for your future responses:

Environment: All code must be optimized for Google Colab. Include necessary library imports.

Modular Structure: Divide the code into logical Cells (e.g., # Cell 1: Imports, # Cell 2: Data Loading, # Cell 3: Processing).

Visualization Excellence: Create professional, 'presentation-ready' charts. Ensure all axes are labelled, and a clear title is provided.

Error-Free & Robust: Use clean, vectorized Python (Pandas/NumPy). Include brief try-except blocks or data validation checks where appropriate to ensure the code runs without errors.

Documentation: Provide detailed comments in the code that explain the 'why' behind complex logic.

Insights: After each code block, provide 2-3 bullet points of high-level analytical insights derived from the expected output. Do not perform any analysis yet.

Screenshot:



tips (2) (1) (1) (1) (1).csv
Spreadsheet

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Task 2 – Choose 5 important columns & perform Basic Statistics (Mean, Median, Mode, Std Dev, Min, Max)

Prompt:

Task 2:

Choose following columns in the dataset as important: total_bill, tip, sex, smoker, day, time, size.

Note: All further analysis will be done on these columns only.

Please do the descriptive analysis on these columns.

Screenshot:

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total_bill, tip, sex, smoker, day, time, size.

Note: All further analysis will be done on these columns only.
Please do the descriptive analysis on these columns.

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Here's a clean, **Google Colab-ready** solution for Task 2 with modular cells, robust checks, and presentation-quality outputs.

◆ # Cell 1: Imports

<> Python



Run

Cell 1: Import required libraries

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt



Set display options for better readability

pd.set_option('display.max_columns', None)

Task 3 – Check for missing values

Prompt:

Task 3: Check for missing values in the specified columns to validate data quality. Note: The required code has already been executed in this Google Colab notebook, so please avoid repeating it.

Screenshot:

Task 3: Check for missing values in the specified columns to validate data quality.
Note: The required code has already been executed in this Google Colab notebook, so please avoid repeating it.

Since the dataset is already loaded and the required columns (`df_selected`) are prepared, we will **only run** the missing values check without repeating previous steps.

◆ # Cell: Missing Values Validation (Incremental)

Python



Run

```
# Missing Values Check (Incremental - No Repetition)
```

```
try:
```

```
    # Compute missing values count and percentage
```

```
    missing_count = df_selected.isnull().sum()
```

```
    missing_percent = (missing_count / len(df_selected)) * 100
```

```
    # Combine into a single DataFrame for better readability
```

Task 4 – Check for Outliers:

Prompt:

Task 4: Perform outlier detection using appropriate graphical methods to visually assess the presence and distribution of extreme values in the dataset.

Screenshot:

Task 4: Perform outlier detection using appropriate graphical methods to visually assess the presence and distribution of extreme values in the dataset.

Great — now we'll perform **outlier detection using high-impact visualizations** without repeating earlier steps.

We'll use:

- **Boxplots** → Best for detecting outliers
- **Histograms** → To understand distribution & skewness

◆ # Cell: Outlier Detection using Boxplots

Task 5 – Univariate Analysis (Visualize the distribution of a column)

Prompt:

Task 5:

Conduct a comprehensive univariate analysis with mentioned columns of the dataset emphasizing on visual storytelling, using appropriate plots to clearly communicate the distribution, central tendency, variability, and key patterns within each variable.

Screenshot:

Task 5:

Conduct a comprehensive univariate analysis with mentioned columns of the dataset emphasizing on visual storytelling, using appropriate plots to clearly communicate the distribution, central tendency, variability, and key patterns within each variable.

Excellent — now we'll perform a **comprehensive univariate analysis with strong visual storytelling**, focusing on both **numerical** and **categorical** variables.

We'll use:

- **Histograms + KDE** → Distribution & skewness
- **Boxplots** → Spread & outliers
- **Bar charts** → Categorical distribution

◆ # Cell 1: Univariate Analysis – Numerical Variables

Task 6 – Bi-variate Analysis (Relationship between main column (Survived/ Tips) and other 4 important columns one-by-one)

Prompt:

Task 6: Perform a bivariate analysis to examine the relationship between the primary target variable (e.g., Tips) and each of the four selected key variables individually, using appropriate statistical summaries and visualizations

Screenshot:

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Great — now we move into **bivariate analysis**, focusing on how the **target variable** (`tip`) relates to other key variables:

- `total_bill`
- `size`
- `sex`
- `smoker`

We'll combine **statistical summaries** + **high-impact visualizations** for clear storytelling.

◆ # Cell 1: Tip vs Total Bill (Continuous v  Continuous)

Final Takeaways

1. `Total_bill` shows the strongest relationship with `tip`.
2. `Size` influences `tip` but with higher variability.
3. `Sex` and `smoker` show behavioral differences, often subtle.